

Material UI Day1 Lab Assignments

By the end of this lab, you should be able to:

- Understand and implement Data Display Components in MUI.
 - Use Input Components to capture user interactions.
 - Customize Material UI components using props and styles.
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Section 1: Data Display Components

Task 1: Typography

Instructions:

1. Create a React Component that displays the following:
 - A main heading (h1) with the text "Material UI Lab Assignments".
 - A subheading (h4) with the text "Day 1: Data Display and Inputs".
 2. Use different variant props (h1, h4, subtitle1,..) for text styling.
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Task 2: Icons (Material Icons)

Instructions:

1. Install Material Icons if not already installed:
`npm install @mui/icons-material`
 2. Create a component that displays the following icons:
 - Home icon with size large and color primary.
 - Favorite (heart) icon with color error.
 - Settings icon with size small.
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Task 3: Avatar

Instructions:

1. Create a User Profile Card that contains:
 - An Avatar displaying a user image from a URL.
 - An Avatar with only initials (e.g., "JS" for John Smith).
 2. Use different variant props (circle, rounded, square).
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Task 4: Chip

Instructions:

1. Create a component that displays different Chips:
 - A simple chip with label "Basic Chip".
 - A clickable chip that logs a message when clicked.
 - A deletable chip with an onDelete function.
 - A chip with an Avatar.
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Section 2: Input Components

Task 5: Buttons

Instructions:

1. Create different Button Variants:
 - A Contained button with the text "Click Me".
 - An Outlined button with text "Outlined Button".
 - A Text button.
 2. Add an onClick event that displays an alert when clicked.
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Task 6: Button Group

Instructions:

1. Create a Button Group containing three buttons:
 - "Option 1"
 - "Option 2"
 - "Option 3"
 2. The group should have contained styling.
 3. Align the buttons horizontally.
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Task 7: Checkbox

Instructions:

1. Create a checkbox with the label "Accept Terms and Conditions".
 2. Implement an onChange event that logs whether the checkbox is checked.
 3. Add an indeterminate checkbox.
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Task 8: Floating Action Button (FAB)

Instructions:

1. Create a Floating Action Button with an Add (+) icon.
 2. Position it at the bottom-right corner of the page.
 3. Make the button change color when hovered.
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Task 9: Radio Group

Instructions:

1. Create a Radio Group with two options: Male and Female.
2. Display the selected value in a heading when a radio button is selected.

Task 10: Rating

Instructions:

1. Create a 5-star Rating System.
 2. Allow users to select ratings with 0.5 precision.
 3. Display the selected rating value below the stars.
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Task 11: Select (Dropdown)

Instructions:

1. Create a dropdown (Select) with the following options:
 - "React"
 - "Angular"
 - "Vue"
 2. Display the selected value in a paragraph below the dropdown.
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Task 12 : Text Field

Instructions:

1. Create a Text Field with the label "Enter your name".
 2. Add an event that displays the typed value in a heading.
 3. Enable full-width styling.
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Task 13: Toggle Button

Instructions:

1. Create a set of Toggle Buttons for selecting a theme:

- "Light Mode"
- "Dark Mode"

2. When a button is selected, change the background color accordingly.